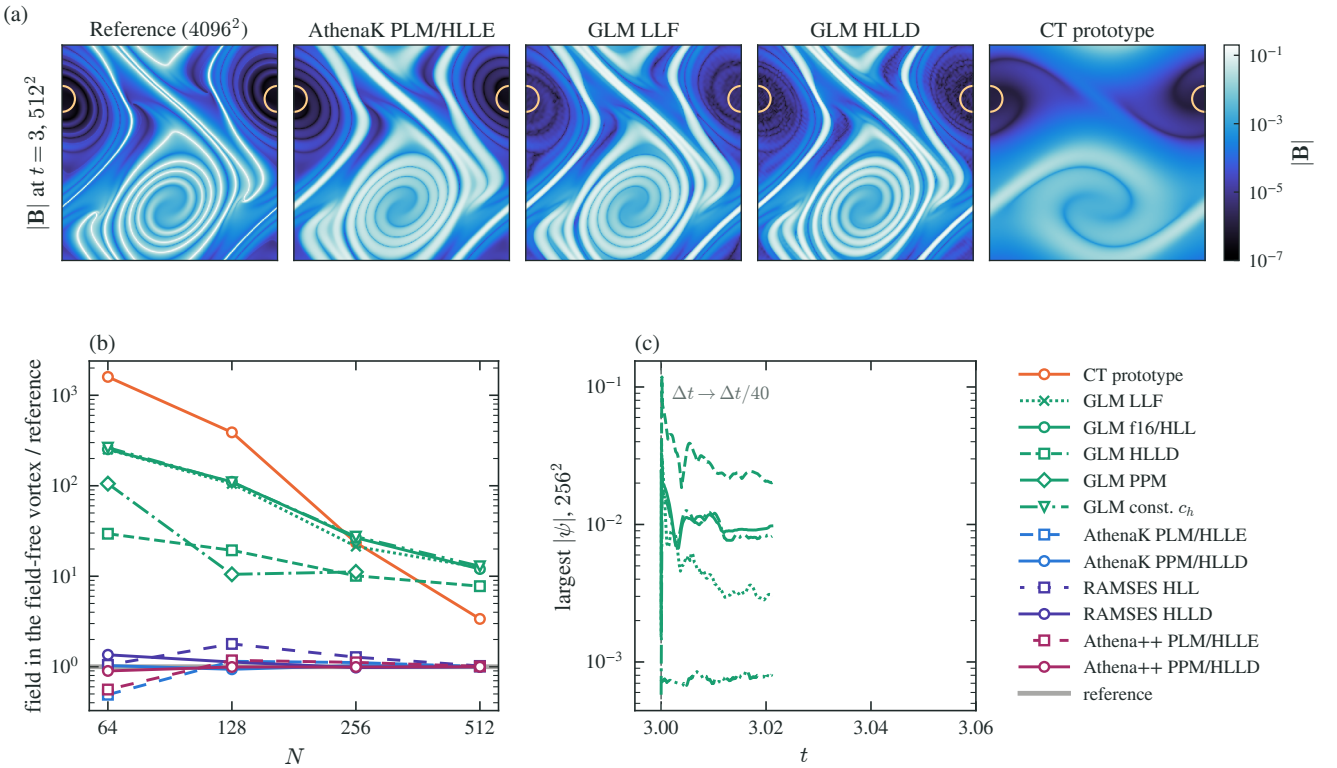

(a)
Reference (4096^2)
AthenaK PLM/HLLE
GLM LLF
GLM HLLD
CT prototype
$|\mathbf{B}|$ at $t = 3,\ 512^2$
10^{-1}
10^{-3}
10^{-5}
10^{-7}
$|\mathbf{B}|$
(b)
field in the field-free vortex / reference
10^3
10^2
10^1
10^0
64
128
256
512
N
(c)
largest $|\psi|$, 256^2
$\Delta t \to \Delta t/40$
10^{-1}
10^{-2}
10^{-3}
3.00
3.02
3.04
3.06
t
CT prototype
GLM LLF
GLM f16/HLL
GLM HLLD
GLM PPM
GLM const. c_h
AthenaK PLM/HLLE
AthenaK PPM/HLLD
RAMSES HLL
RAMSES HLLD
Athena++ PLM/HLLE
Athena++ PPM/HLLD
reference